Project Proposal: Targeted Indoor Aldehyde Remediation and Lifetime Cancer Risk Mitigation

1. Executive Summary & Problem Statement

Volatile organic compounds (VOCs)—specifically carcinogenic low-molecular-weight aldehydes such as formaldehyde, acetaldehyde, and acrolein—represent one of the most pervasive yet unaddressed indoor air quality (IAQ) hazards in modern urban environments. Standard mechanical filtration technology, such as High-Efficiency Particulate Air (HEPA) filters, is engineered exclusively to capture suspended particulate matter and fails entirely against gaseous chemical pollutants. Meanwhile, conventional sorbents like activated carbon rely on weak physical adsorption (physisorption) governed by van der Waals forces. Because light aldehydes possess high vapor pressures, activated carbon suffers from rapid ambient desorption, releasing trapped toxins back into indoor spaces during temperature or humidity fluctuations.

The resulting chronic inhalation exposure contributes significantly to cumulative excess lifetime cancer risks, particularly in urban environments where structural off-gassing, modern composite building materials, and traffic-dense commuting corridors compound daily doses. This proposal details a high-efficiency, low-pressure-drop drop-in filter attachment utilizing an electrospun nanofiber matrix (OEM) functionalized with amine-bearing metal-organic frameworks (MOFs) alongside alternative catalytic and chemical capture vectors. Designed as an interdisciplinary project suitable for institutional incubation, this technology aims to achieve permanent chemical sequestration or degradation of airborne aldehydes at the molecular level.

2. Summary of Comprehensive Cancer Risk Multipliers Across Urban Environments

To evaluate the clinical and public health necessity of this filtration technology, cumulative lifetime excess cancer risks were modeled across major American urban environments under standardized 24-hour time-activity schedules (allocating time across work, home, vehicle, and outdoor microenvironments). Risk factors are expressed as multiples of the EPA reference concentration baseline.

Single-Chemical Formaldehyde Risk Multipliers

- **Low-Income Environments:** Driven by denser traffic corridors, older building materials, and smaller living spaces, low-income baseline exposures range from 1.71 times up to 8.38 to 9.14 times under old building heat off-gassing conditions. New building workplace exposure consistently yields 5.43 to 6.64 times baseline risk.
- **Middle-Income Environments:** Standardizing home and vehicle baselines while evaluating workplace variables yields overall risk multipliers ranging from 0.98 times to a

peak of 7.20 to 7.82 times under old building off-gassing conditions. Normal indoor and new building workplaces generate sustained risks between 2.44 and 5.32 times.
- **High-Income Environments:** Benefiting from better ventilated housing stock and shorter commutes, high-income baseline risks span from 0.57 times up to a peak of 6.79 to 7.20 times when exposed to old building thermal off-gassing in the workplace.

Combined Aldehyde Cumulative Risk Multipliers When incorporating additive co-carcinogens and upper-respiratory mutagens (specifically acetaldehyde from combustion exhaust and acrolein), the total multi-aldehyde toxicity load increases the effective risk multiplier across all microenvironments:

- **Low-Income Combined Risk:** Ranges from 2.31 times to peak cumulative exposures of 10.71 to 12.34 times under old building thermal off-gassing conditions. New building workplaces drive combined risks between 7.33 and 8.96 times baseline.
- **Middle-Income Combined Risk:** Spans from 1.32 times up to a peak of 9.72 to 10.56 times in old building off-gassing workplaces. Standard new building office environments produce combined risk multipliers between 6.35 and 7.18 times.
- **High-Income Combined Risk:** Ranges from 0.77 times to peak old building off-gassing risks of 9.17 to 9.72 times. Standard new building workplaces maintain combined risk levels between 5.79 and 6.35 times baseline.

Health Outcomes and Evidence Table

Cancer Type	Outcome	Latency	Prognosis (5-yr survival)
Nasopharyngeal carcinoma	IARC Group 1 (sufficient)	15-30 yr	Localized ~85%, regional ~73%, distant ~49%; overall ~62%; chemoradiation-responsive
Myeloid leukemia / MDS	Strong (recent NCI-cohort reanalyses)	2-15 yr	AML: ~55-60% (age<45), ~30% (45-64), ~10% (65+); MDS by IPSS-R (>8 yr to ~1.3 yr median)
Sinonasal carcinoma	IARC Group 1	20-30 yr	~55-60% overall (~85% localized)
Non-cancer: asthma, sensitization, chronic rhinitis	Strong	0-5 yr (reversible early)	Persistent lung-function loss if exposure continues

3. Technical Mechanism: Potential Solutions

To address the limitations of passive physisorption, we examine multiple chemical and catalytic mechanisms integrated into high-porosity electrospun polymer backbones.

Solution Vector (Prime): Amine-MOF Functionalized Nanofiber Matrix (OEM)

- **Substrate Fabrication:** The matrix is electrospun from polymer solutions (such as polyacrylonitrile or polyvinylidene fluoride), yielding an interconnected web of sub-micron fibers ranging from 100 to 500 nanometers in diameter. This open-pore geometry minimizes aerodynamic pressure drop while maximizing gas-to-surface contact time.
- **Targeted Chemisorption:** Zirconium- or chromium-based MOF crystals with internal surface areas exceeding 1,000 square meters per gram are anchored within the nanofibers.
- **Covalent Binding:** Primary and secondary amine groups grafted onto the MOF linkers perform a nucleophilic attack on the carbonyl carbon of diffuse aldehydes. Condensation forms a stable imine (Schiff base) covalent bond, preventing ambient desorption.

Alternative Solution Vector 1: Base-Catalyzed Disproportionation (Cannizzaro Pathway)

- **Mechanism:** Non-enolizable aldehydes (specifically formaldehyde) undergo base-catalyzed disproportionation where two formaldehyde molecules react with a hydroxide ion to yield one molecule of methanol and one molecule of formate.
- **Implementation:** Alkaline earth oxides (such as calcium oxide or magnesium oxide) or strong basic anion-exchange resin nanoparticles are embedded directly into the electrospun web.
- **Trade-off Analysis:** Offers continuous catalytic conversion for formaldehyde without relying on stoichiometric organic amine consumption. However, enolizable aldehydes such as acetaldehyde and acrolein undergo competing self-condensation (aldol) side reactions rather than pure Cannizzaro conversion, requiring composite catalytic formulations.

Alternative Solution Vector 2: Ambient Photocatalytic & Thermocatalytic Oxidation (PCO)

- **Mechanism:** Transition-metal oxides (such as manganese oxide or cerium oxide) or single-atom noble metals (such as platinum supported on titanium dioxide) catalyze the full oxidation of low-molecular-weight aldehydes in the presence of oxygen, yielding carbon dioxide and water molecules at room temperature.
- **Implementation:** Active catalytic nanoparticles are surface-coated onto high-surface-area electrospun nanofiber beds.
- **Trade-off Analysis:** Provides theoretical infinite operational capacity (zero chemical saturation). However, efficacy can be compromised by competitive adsorption with humidity under variable real-world indoor ambient conditions.

Alternative Solution Vector 4: High-Loft Web of Amine-MOF (“Cotton Candy”) Sandwiched Between Semirigid Scrims

- **Mechanism:** High-loft, voluminous polymer micro-fibers are expanded via solution-blowing, melt-blowing, or air-laid processes into an open-celled 3D cushion carrying amine-functionalized metal-organic framework (MOF) crystals (such as $UiO - 66 - NH_2$ or ZIF derivatives) throughout its depth. Primary and secondary amine

groups grafted onto the MOF linkers perform a nucleophilic attack on the carbonyl carbon of diffuse low-molecular-weight aldehydes, forming stable covalent imine (Schiff base) bonds that prevent moisture-driven desorption.

- **Implementation:** The lofted active sorbent core is physically encapsulated between two water-insoluble, semirigid structural non-woven scrims (or support grids) using thermal point-bonding or breathable web adhesives. Hydrophobic polymer backbones—such as **Polyvinylidene Fluoride (PVDF)**, **Polyacrylonitrile (PAN)**, or **Thermoplastic Polyurethane (TPU)**—and hydrolytically stable Zr-based MOFs ensure the core matrix remains completely insoluble and structurally intact when exposed to condensed water or high ambient humidity.
- **Trade-off Analysis:** Maximizes 3D gas contact area and delivers exceptional breathability with ultra-low aerodynamic pressure drop (ΔP), preserving HVAC blower efficiency while eliminating delicate nanofiber handling issues. However, unlike infinite catalytic oxidation, the amine-MOF active sites are stoichiometric and undergo eventual chemical saturation, requiring consumable replacement cycles once spent.

Alternative Solution Vector 4:
Alternative Solution Vector 5:

4. Future Directions

Rather than manufacturing standalone consumer hardware, the venture operates as an advanced materials supplier, licensing formulations and supplying integrated filtration components directly to commercial HVAC providers, cleanroom integrators, and institutional facilities. Because active binding or catalytic capacity on the selected substrate eventually degrades or saturates during continuous aldehyde remediation, if executed as a business model, it would rely on a high-margin consumable replacement cycle—delivering sustainable, recurring revenues as spent filtration media are regularly swapped during scheduled maintenance.

Phased Project Roadmap

- **Step 1 (Campus Incubation & Living Labs):** Deploy initial prototypes within university buildings and living laboratories, utilizing campus facilities funding to establish baseline performance metrics under real-world occupancy loads.
- **Step 2 (Commercial Real Estate & Industrial Pilots):** Target commercial office towers, healthcare facilities, and manufacturing plants facing indoor air quality compliance and WELL building certifications.
- **Further steps depend on success in the first two phases.**

Ideal Estimated Ultimate Impact: Campus-Scale Phase 1 Deployment

Restricting initial deployment to Phase 1 within targeted university campus facilities establishes a localized, highly controlled proof-of-concept environment that yields immediate scientific and operational impact:

- **Direct Toxicological Risk Reduction:** Concentrating initial installations within designated high-risk campus microenvironments—such as older administrative buildings prone to thermal off-gassing, research laboratories, and high-density student residential halls—directly mitigates peak aldehyde exposure spikes. This provides measurable indoor air quality remediation for localized campus populations, driving exposure levels down toward baseline thresholds.
- **Real-World Empirical Validation:** Operating as a campus "living laboratory" provides long-term operational data across real-world occupancy loads, relative humidity fluctuations, and variable airflow profiles. This empirical validation establishes exact performance degradation curves, pressure drop profiles, and saturation timelines for each competing candidate technology to become broadly available for commercial use.

5. Stake Holders

Potential stakeholders were selected and categorized based on the alignment of their technical expertise and academic discipline with the primary engineering and commercialization challenges of the project. Core faculty advisors and campus facilities managers were engaged to evaluate structural integration, indoor air quality compliance, and living-lab deployment logistics within institutional infrastructure. Simultaneously, academic researchers and industry mentors specializing in chemical engineering, materials science, and biomedical disciplines were paired with student teams based on majors for 'areas' of expertise. The following is a temporary sorting,

1. Shohom (BME) **Shohom_Stakeholder_Email_Drafts_GCLLC'26**

Focus: High-impact MOF chemistry advisors, solution-side technical leads, regulatory bodies, medical exposure environments, and primary filtration industry leaders.

1. ~~**Dr. Krista Walton (GT ChBE) –** *High-Value Material Contact (SOLUTION)*~~
    - ~~**Task:** Discuss selecting amine-functionalized MOFs and thermodynamic selectivity for nucleophilic aldehyde capture.~~
2. ~~**Dr. Christopher Jones (GT ChBE) –** *High-Value Chemistry Contact (SOLUTION)*~~
    - ~~**Task:** Consult on sorbent amine chemistry and aldehyde-Schiff base reactions on solid substrates.~~
3. **Donaldson Company** – *Primary Industry Contact*
    - **Task:** Interview on current commercial VOC/aldehyde filtration technologies, manufacturing capabilities, and industry awareness of off-gassing exposure gaps.
4. **Freudenberg Filtration Technologies** – *Primary Industry Contact*
    - **Task:** Interview on specialty HVAC filter media, active sorbent multi-layer inserts, and current approaches to gaseous aldehyde capture.
5. **US EPA – Center for Public Health & Environmental Assessment (CPHEA)**
    - **Task:** Validate IRIS toxicity benchmarks, 24-hr TWA targets, and cancer risk scaling parameters.
6. **NIOSH – Indoor Environment Group**
    - **Task:** Scope occupational exposure limits (RELs) and standard gas sampling protocols.
7. **Emory University Environmental Health & Safety (EHS)**
    - **Task:** Evaluate acute formaldehyde off-gassing and exposure standards in histology/pathology labs.
8. **Piedmont Healthcare Facilities Engineering**
    - **Task:** Assess hospital disinfectant/sterilant off-gas capture requirements.

2. Aegean (MSE)

Focus: Construction materials off-gassing, polymer binder hydrolysis, synthetic resin mechanics, and building product emissions.

1. **UL Solutions / GREENGUARD Certification Group (Atlanta Office)**
    - **Task:** Scope low-emitting product standards, chamber off-gassing test protocols, and VOC certification limits for building materials.
2. **Resin & Adhesive Technical Specialists at Georgia-Pacific (Atlanta, GA)**
    - **Task:** Investigate urea-formaldehyde (UF) and phenol-formaldehyde (PF) resin hydrolysis, off-gassing dynamics, and emissions over time.
3. **Shaw Industries Technical Center (Dalton, GA)**
    - **Task:** Gather field data on carpet backing off-gassing, synthetic binders, and VOC emission decay curves in interior products.
4. **Dr. Preet Singh (GT MSE)**
    - **Task:** Consult on environmental degradation pathways of polymers and building materials when exposed to ambient moisture and heat.
5. **American Wood Council (AWC) – Environmental & Health Division**
    - **Task:** Understand engineered wood product off-gassing characteristics (plywood, particleboard, MDF) and industry baseline levels.
6. **Composite Panel Association (CPA)**
    - **Task:** Gain insights on formaldehyde emissions testing standards (TSCA Title VI / CARB Phase 2) for composite wood products.

3. Ernest (MechE)

Focus: Building ventilation dynamics, commercial HVAC air exchange, construction mechanical systems, and real estate envelope seals.

1. **ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) – Technical Committee 2.4 (Gaseous Contaminants)**
    - **Task:** Consult on standard ventilation rates (62.1/62.2), air cleaner testing standards (Method 145.2 for gas-phase contaminants), and HVAC pressure drop allowances.
2. **Acuity Brands Building Management Group (Atlanta, GA)**
    - **Task:** Assess building automation systems, airflow circulation constraints, and commercial HVAC retrofit feasibility.
3. **Cousins Properties (Atlanta Real Estate Operations)**
    - **Task:** Scope mechanical HVAC constraints, filter frame tolerances, and pressure drop limits in high-rise commercial builds.
4. **Dr. Marta Hatzell (GT ME / ChBE)**
    - **Task:** Analyze airflow fluid transport, ductwork velocity profiles, and mechanical integration of secondary inserts.
5. **U.S. Green Building Council (USGBC) – LEED Indoor Environmental Quality Group**

    - **Task:** Understand LEED ventilation requirements, flush-out protocols for new construction off-gassing, and IAQ testing frameworks.
6. **Southface Institute Engineering Team (Atlanta, GA)**
    - **Task:** Scope building envelope airtightness, mechanical ventilation deficits in residential builds, and field IAQ monitoring equipment.

4. Derek (Aerospace)

Focus: Gas-phase kinetics, vapor diffusion modeling, regional atmospheric off-gassing chemistry, and regulatory monitoring.

1. **Dr. David Sholl (GT ChBE / Oak Ridge Nat'l Lab)**
    - **Task:** Discuss computational modeling of gas-phase transport, concentration gradients, and vapor diffusion through porous matrices.
2. **Dr. Ryan Lively (GT ChBE)**
    - **Task:** Analyze gas transport dynamics, mass transfer coefficients, and pore diffusion in high-velocity airflow streams.
3. **OSHA – Office of Chemical Hazards**
    - **Task:** Scope industrial compliance standards, Permissible Exposure Limits (PELs), and enforcement thresholds for volatile organic compounds.
4. **Georgia Department of Public Health (DPH) – Environmental Health Section**
    - **Task:** Gather data on regional indoor air quality monitoring, off-gassing exposure hotspots, and community health interventions.
5. **Air & Waste Management Association (A&WMA) – Indoor Environmental Quality Committee**
    - **Task:** Consult on field measurement techniques for low-concentration gaseous aldehydes in ambient and indoor air.
6. **Georgia Tech IEN Cleanroom / Characterization Staff**
    - **Task:** Inquire about gas sampling equipment, mass spectrometers, and analytical setups for measuring gas-phase concentrations.

5. Joanna (BME)

Focus: Indoor air toxicology, residential building off-gassing impacts, occupant exposure pathways, and housing infrastructure.

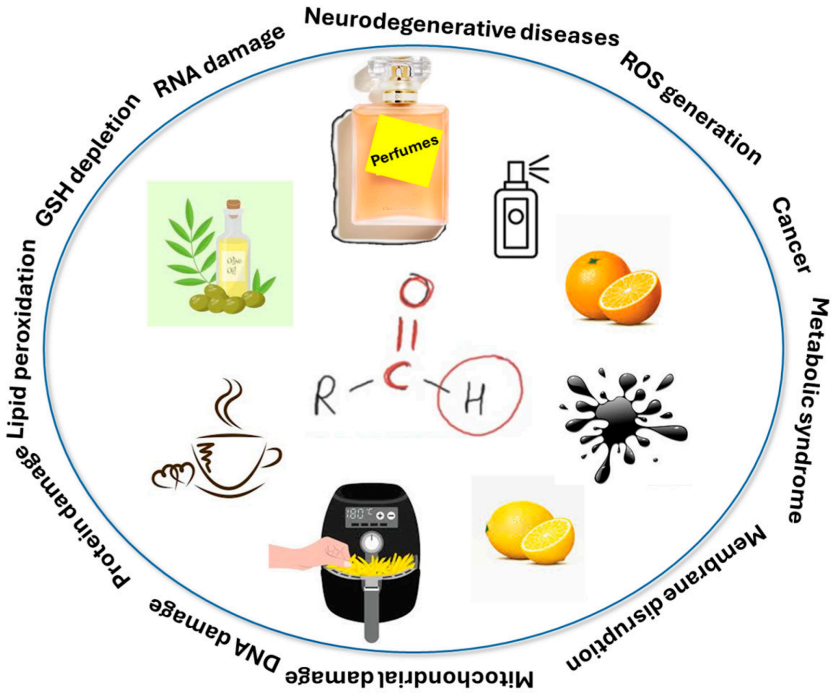
1. **Indoor Air Quality Association (IAQA) – Health & Toxicology Chapter**
    - **Task:** Scope health outcomes associated with chronic sub-ppm formaldehyde off-gassing in residential vs. commercial settings.
2. **Atlanta Housing Authority (AHA) – Facilities & Environmental Management**
    - **Task:** Validate residential indoor air quality issues, poor ventilation factors, and high-density off-gassing in public housing.
3. **National Environmental Health Association (NEHA)**
    - **Task:** Gather data on indoor exposure protocols, occupant risk communication, and residential off-gassing assessments.

4. **Dr. Paul Kohl (GT ChBE)**
    - **Task:** Consult on reactive gas interactions with interior synthetic surfaces and long-term material off-gassing chemistry.
5. **Dr. Meisha Shofner (GT MSE)**
    - **Task:** Discuss structural stability of porous substrate sheets and physical property changes during exposure to ambient humidity and volatile vapors.
6. **GT Environmental Health & Safety (EHS)**
    - **Task:** Scope campus building off-gassing baselines, laboratory safety protocols, and real-time formaldehyde monitoring practices.

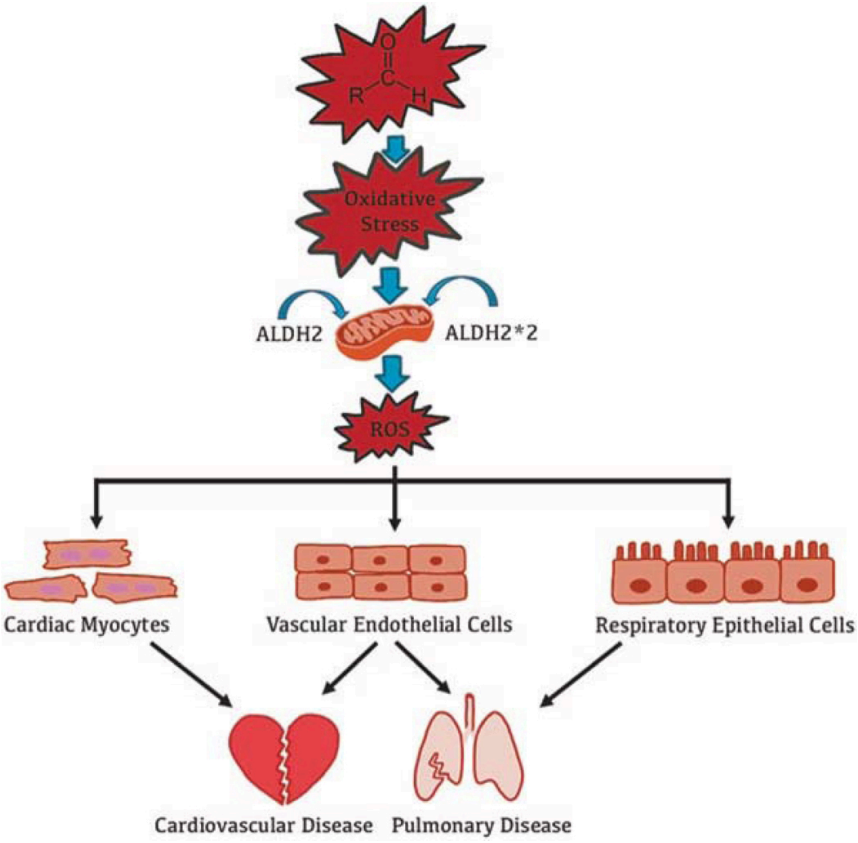
Separate Hold List: Solution Vector & Fabrication Stakeholders

Isolate these contacts for later project phases when prototype design, electrospinning, and MOF layer fabrication begin.

- **Dr. Sundaresan Jayaraman (GT MSE)** – *Functional fibrous materials and polymer electrospinning.*
- **Georgia Tech Renewable Bioproducts Institute (RBI) Fiber Processing Core** – *Scalable web-laying and biomaterial fiber processing.*
- **Dr. Vladimir Golubovic (GT MSE)** – *Polymer matrix synthesis and electrospun fiber degradation chemistry.*
- **Dr. Karl Jacob (GT MSE)** – *Rheology, solution viscosity, and fiber spinning parameters.*
- **AHLSTROM** – *Non-woven media scale-up.*
- **Hollingsworth & Vose** – *Specialty non-woven substrate materials.*



[Sources and toxicity of aldehydes]



[Aldehydes damage the electron transport chain and affect the cardiopulmonary system]
[ALDH2 metabolizes the aldehydes to less toxic form]

ALDH2*2 prevents efficient metabolism of reactive aldehydes → cellular dysfunction and cardiopulmonary disease

1. Secondary Research

Category	Research Area / Focus	Specific Research Questions
Microenvironment Baselines	Unmonitored Indoor Environments	What are the exact 24-hour time-weighted average (TWA) aldehyde concentrations across unmonitored microenvironments such as public transit vehicles, newly renovated multi-family housing, and elementary school classrooms?
	Thermal & Humidity Off-Gassing Cycles	How do real-world diurnal indoor temperature and humidity fluctuations quantitatively alter the rate of thermal off-gassing from composite materials, resins, and synthetic furnishings over multi-year timelines?
Toxicology & Biomarkers	Molecular Biomarkers of Injury	What specific molecular biomarkers (e.g., DNA-protein crosslinks, specific hemoglobin adducts) accurately measure internal effective dose and early-stage cellular damage from chronic sub-ppm indoor formaldehyde exposure?
	Multi-Aldehyde Co-Exposure Interaction	How do low-dose co-exposures to multiple aldehydes (formaldehyde, acetaldehyde, acrolein) interact at the cellular level, and are their cumulative mutagenic and carcinogenic effects strictly additive or synergistic?
Epidemiology & Attribution	Confounder Isolation	How can chronic, low-concentration aldehyde exposure be definitively isolated as a primary causal driver in human epidemiological cohort studies given confounding factors like ambient traffic pollution and $PM_{2.5}$?

	Latency & Population Variances	What are the long-term, real-world latency profiles and regional risk variances for non-nasopharyngeal malignancies (e.g., myeloid leukemia subtypes and sinonasal carcinomas) across different socio-economic urban demographics?
Regulatory & Legal	Toxic Tort Evidentiary Standards	What are the legal evidentiary standards and dose-reconstruction methods required by courts to establish causation in toxic tort litigation involving diffuse, multi-source indoor air off-gassing?
	Institutional Compliance & Policy Gaps	How do indoor air quality compliance gaps between occupational regulatory bodies (OSHA Permissible Exposure Limits) and public health guidance thresholds (EPA IRIS benchmarks) impact institutional liability and policy enforcement?